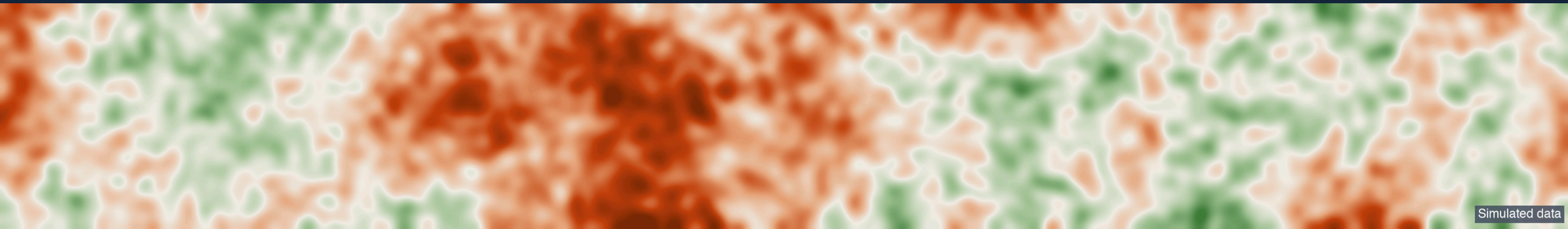
ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Drought Monitoring with Remote Sensing

Watching rain, crops, soil and groundwater dry out from space

AKSHAY RAJ MANOCHA



GOAL OF THIS LECTURE

Use satellite data to tell when a drought starts, how severe it is, where it is worst, and which water supplies it reaches.

Drought types, what satellites measure, the main indices, real droughts, and engineering decisions.

THIS LECTURE

Outline

01 **Drought and why we monitor it**
Types of drought, how it spreads, and what satellites add

02 **What satellites measure**
Rain, vegetation, temperature, soil moisture and groundwater

03 **Drought indices**
SPI, VCI, TCI, VHI and how to combine them

04 **Case studies and services**
Marathwada, the Horn of Africa, California, Europe, and who monitors drought

05 **Engineering practice**
Water decisions, a monitoring workflow, limits and exercises

01

PART 1

Drought and why we monitor it

Types of drought, how it spreads, and what satellites add

DROUGHT

A slow, creeping hazard

A drought is a long period with less water than normal for that place and season. It has no clear start, spreads over huge areas, and its damage builds up for months.

Slow onset

Weeks or months pass before anyone calls it a drought.

Relative

A dry month in Rajasthan is normal; the same month in Kerala is a drought.

Wide areas

Whole states or countries at once, not one valley.

Lasting impacts

Crops, rivers and groundwater can stay low long after the rain returns.

DROUGHT

Four kinds of drought

Meteorological

Rain below normal for weeks to months.

Rainfall: gauges and satellites

Agricultural

Soil too dry for crops; plants are stressed.

Vegetation, temperature, soil moisture

Hydrological

Rivers, lakes and reservoirs fall low.

River flow, reservoir levels and areas

Groundwater

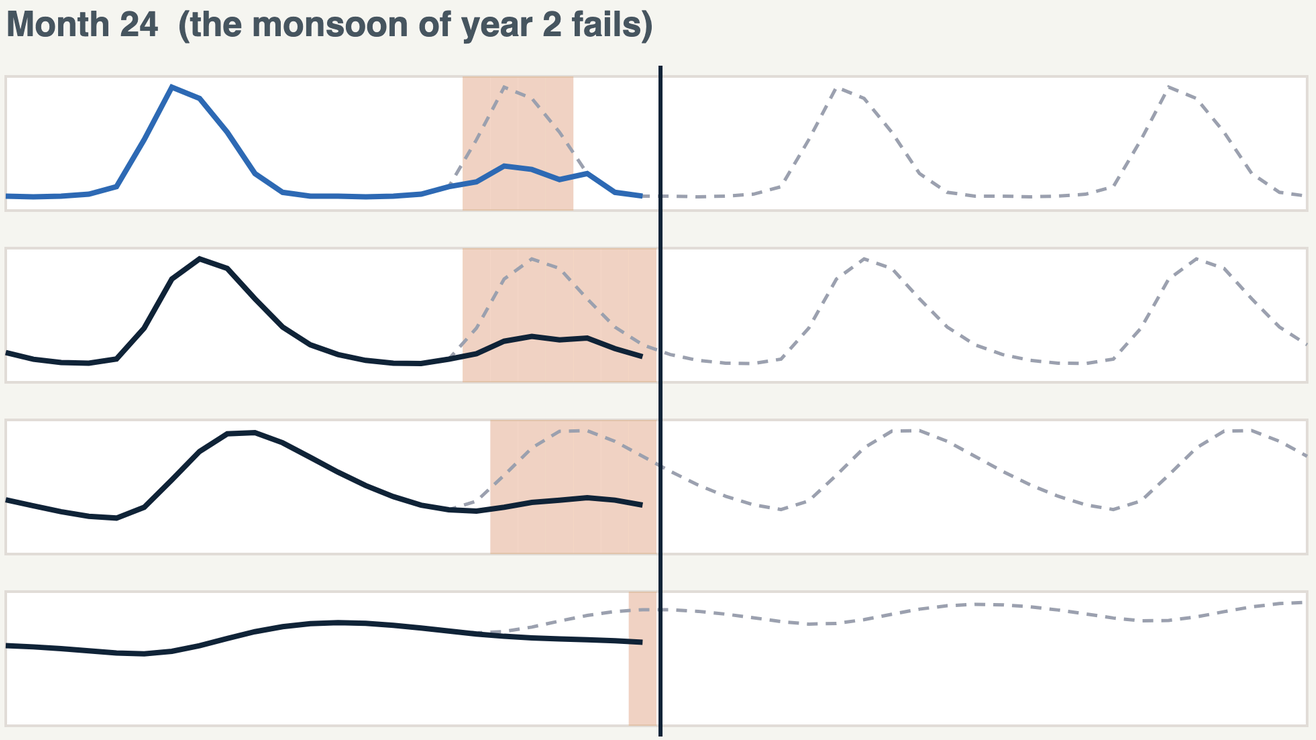
Aquifers drain and wells dry up.

Wells and GRACE gravity satellites

A fifth, socio-economic drought, is when demand for water exceeds the supply people can get.

DROUGHT

How a drought spreads through the water cycle



Dashed: a normal year. Shaded: below 70% of normal.

DROUGHT

Why satellites for drought

Since 1981
Satellite rainfall record (CHIRPS)

Since 2000
MODIS vegetation every 16 days

Since 2002
GRACE water storage, every month

Free
Most drought data and tools

Everywhere at once

Rain gauges and wells are sparse; satellites cover every village, every time.

A long ‘normal’

Decades of consistent images define what is normal for each place and fortnight.

Early warning

Updates every few days catch a drought months before reservoirs run dry.

02

PART 2

What satellites measure

Rain, vegetation, temperature, soil moisture and groundwater

MEASURING

Rainfall from space

CHIRPS

About 5 km, daily to monthly, since 1981; infrared satellite estimates blended with rain gauges.

INSAT-3D / 3DR

Indian geostationary satellites: rainfall estimates every half hour over India.

GPM IMERG

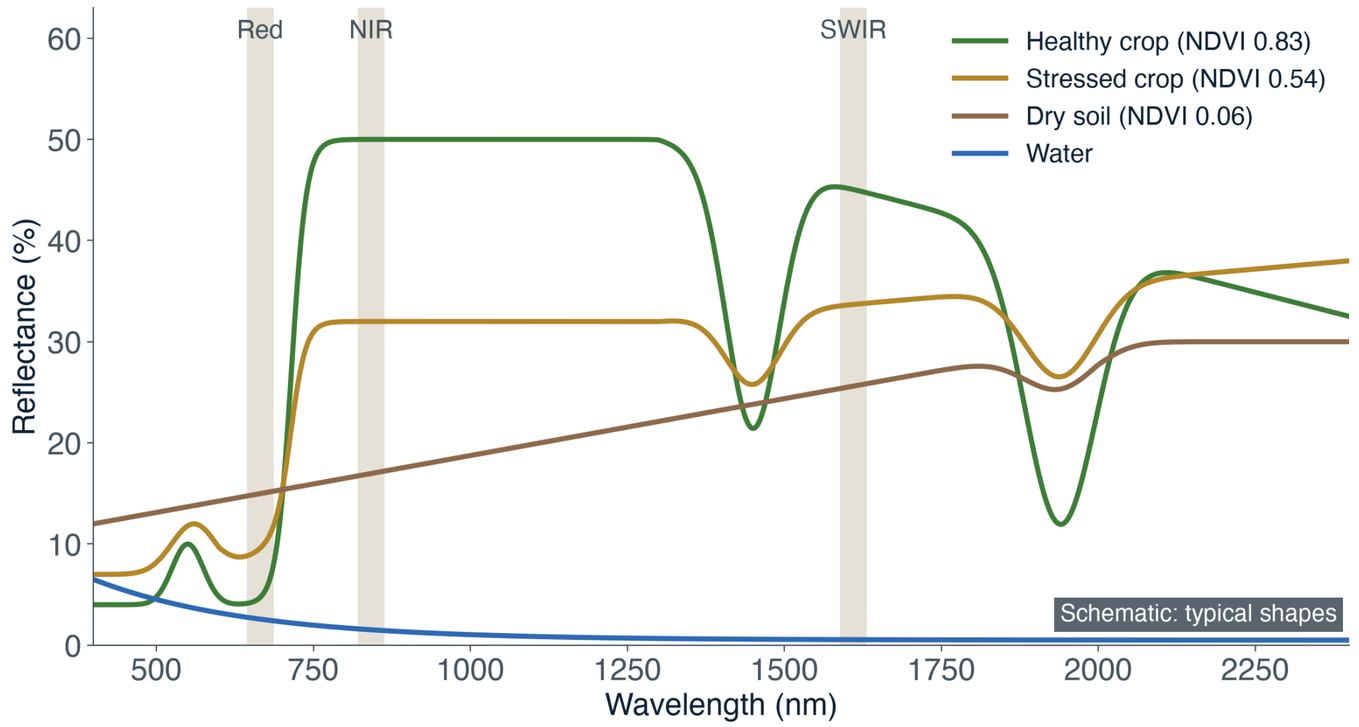
About 10 km, every 30 minutes, from a constellation of microwave satellites.

Then

Rainfall records feed the SPI (Part 3), the standard index for meteorological drought.

MEASURING

Vegetation: how green is the crop?



Schematic: typical reflectance of crops, soil and water.

NDVI = (NIR − Red) ÷ (NIR + Red)

Chlorophyll absorbs red light; healthy leaves reflect near-infrared strongly.

Healthy crop ≈ 0.83, stressed ≈ 0.54, bare soil ≈ 0.06.

Short-wave infrared (SWIR) adds leaf water: indices such as NDWI.

MEASURING

Temperature: a thirsty crop runs hot

Why it works

Plants cool themselves by transpiring. Short of water, they close their pores and warm up.

Land surface temperature

MODIS (1 km, daily), Landsat (100 m), ECOSTRESS (70 m) and INSAT-3D over India.

Evapotranspiration

Energy-balance models turn temperature into water use; the Evaporative Stress Index flags trouble early.

Earlier than NDVI

Temperature rises before leaves turn brown: useful for fast-developing ‘flash’ droughts.

MEASURING

Soil moisture from microwaves

SMAP (NASA)

L-band radiometer since 2015; about 36 km, 9 km products, every 2–3 days.

Sentinel-1 SAR

Radar backscatter rises with soil moisture: about 1 km products, sharper but noisier.

SMOS (ESA)

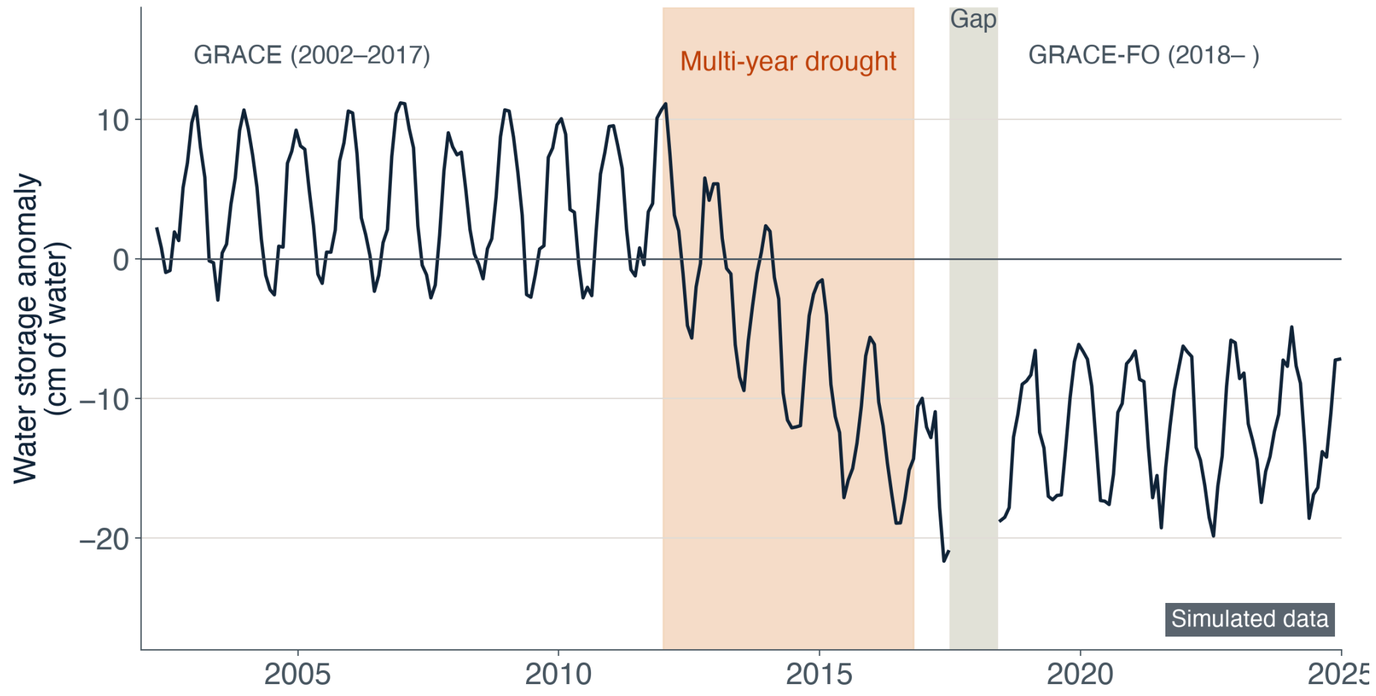
L-band radiometer since 2009; about 40 km.

Only the top few cm

Satellites see the top 5 cm; models extend it down to the root zone.

MEASURING

Groundwater from gravity: GRACE



Simulated: monthly water storage for a large basin.

Two satellites measure tiny changes in the distance between them, caused by changes in Earth’s gravity.

Less water underground means less mass: the total water stored in a region, every month.

Subtract soil moisture and surface water (models) to estimate groundwater.

Coarse: a few hundred kilometres. Basins, not fields.

MEASURING

Which satellite for which part of the drought

What	Satellites	Resolution and record
Rainfall	CHIRPS, GPM IMERG, INSAT-3D	5–10 km; 1981– (CHIRPS)
Vegetation	MODIS, Sentinel-2, Landsat	250 m to 10 m; 2000– (MODIS)
Temperature	MODIS, Landsat, ECOSTRESS	1 km to 70 m
Soil moisture	SMAP, SMOS, Sentinel-1	36 km to 1 km; 2009–
Water storage	GRACE, GRACE-FO	~300 km, monthly; 2002–
Surface water	Sentinel-1 and -2, altimeters	Reservoir area and level

03

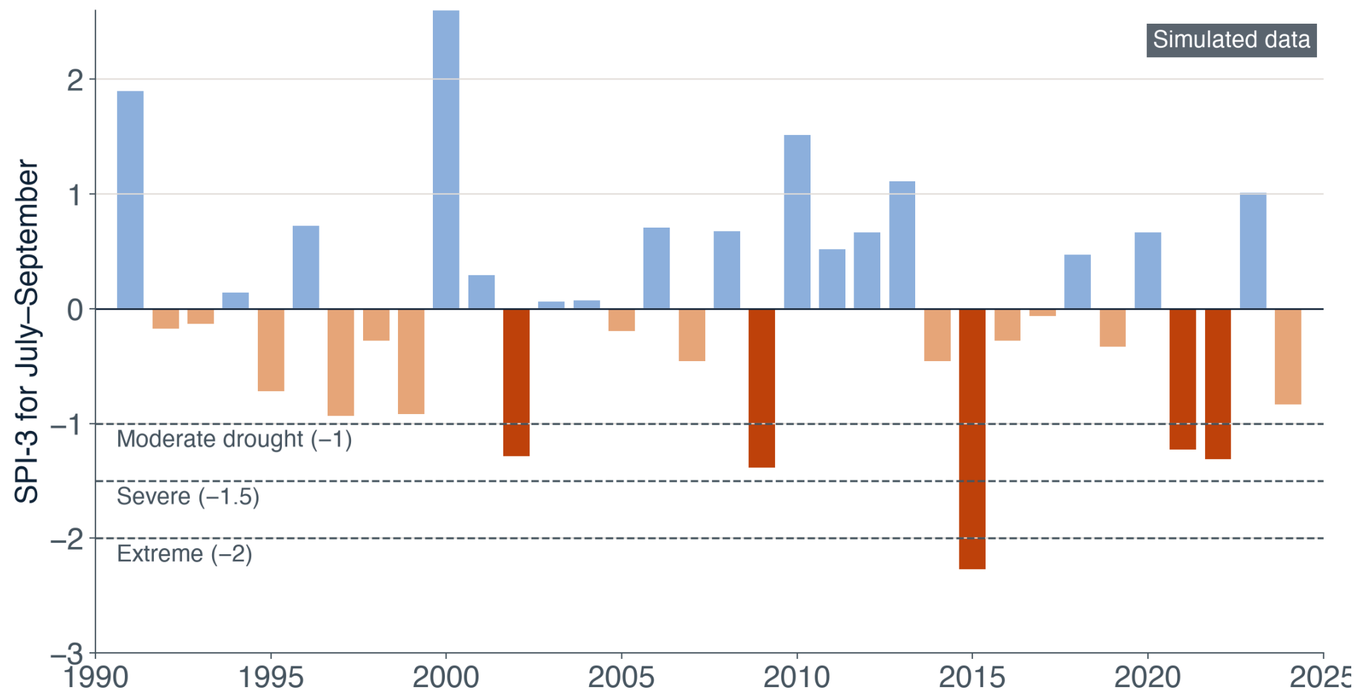
PART 3

Drought indices

SPI, VCI, TCI, VHI and how to combine them

INDICES

SPI: how unusual is the rain?



Simulated monsoon rainfall, 1991–2024.

Fit a probability curve to decades of rainfall for that season, then convert to a standard score.

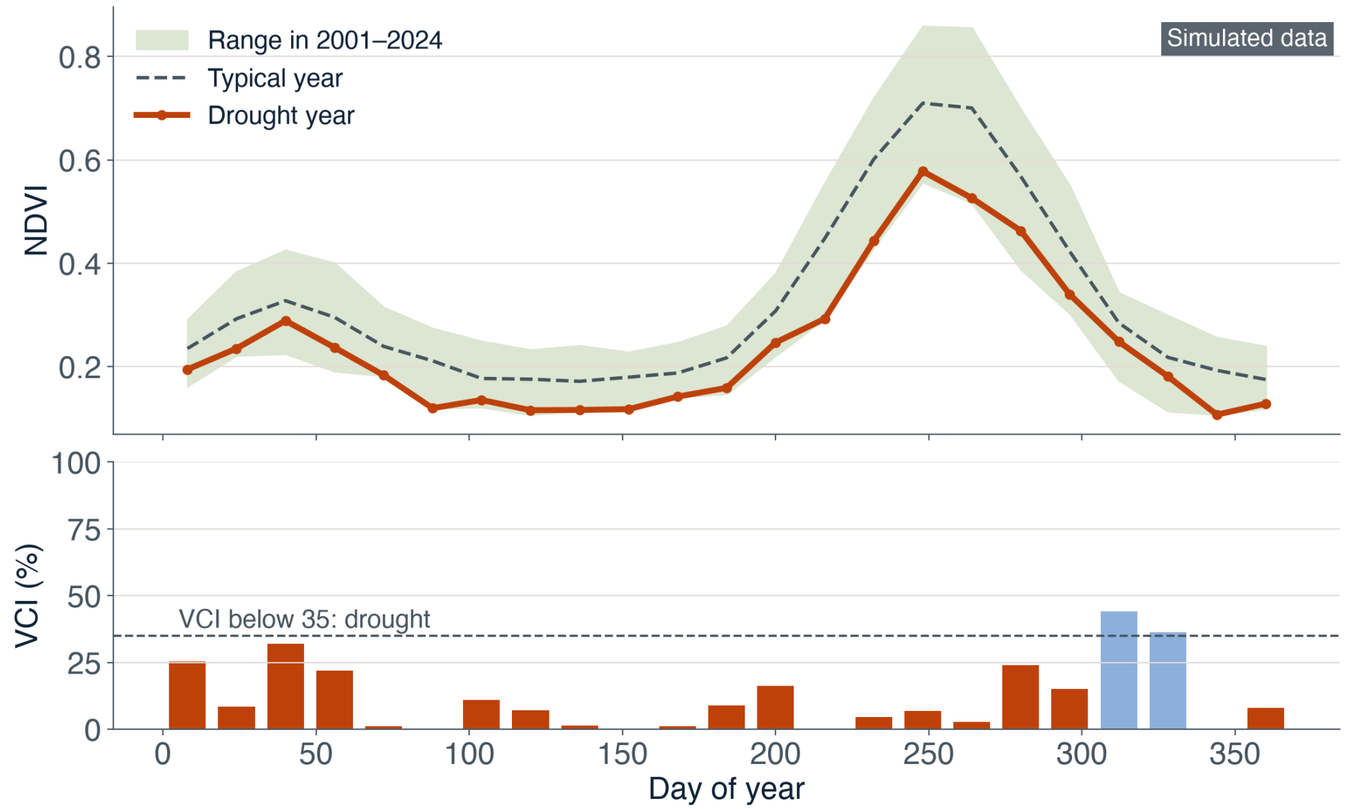
0 is normal; −1 moderate, −1.5 severe, −2 extreme drought (McKee et al., 1993).

Short windows (1–3 months) for crops; 12 months or more for rivers and groundwater.

SPEI adds temperature through evaporation.

INDICES

VCI: greenness against its own history



Simulated crop pixel, 16-day composites.

VCI = (NDVI − min) ÷ (max − min) × 100

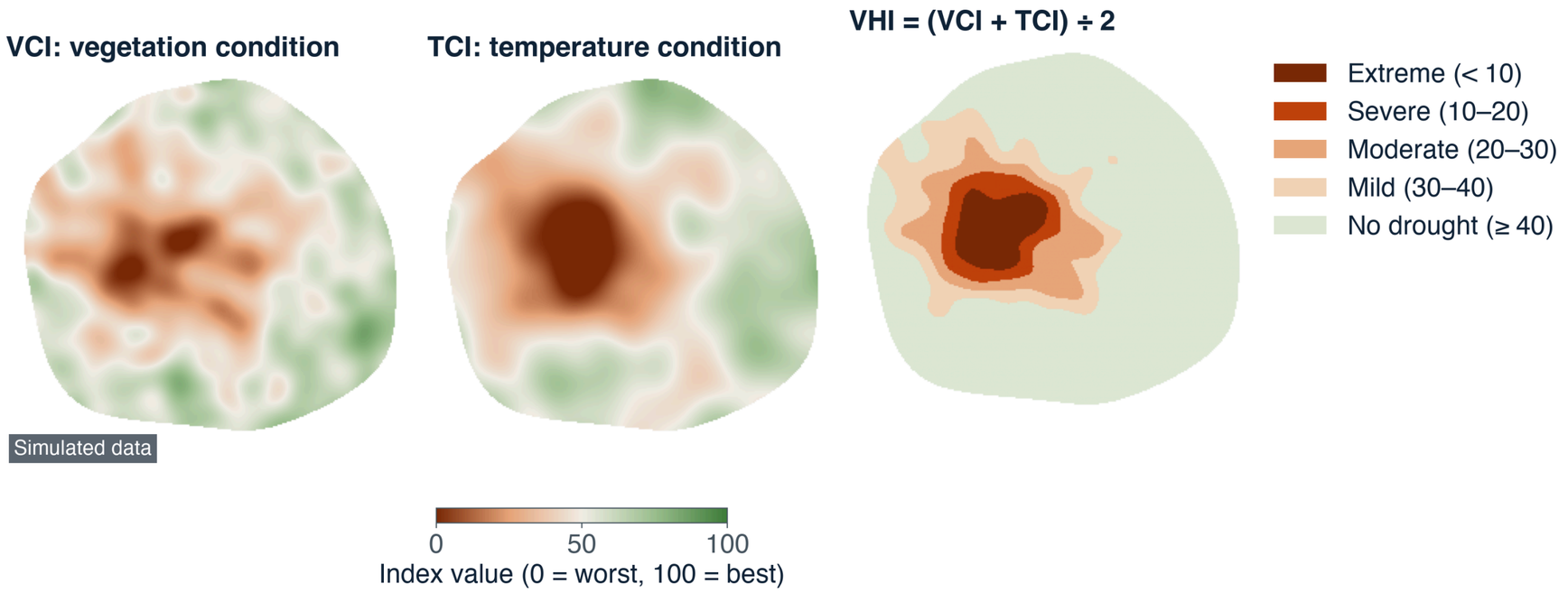
Min and max come from all years for the same fortnight and pixel (Kogan, 1995).

Each place is compared with itself, so desert and forest can share one scale.

A common threshold: VCI below about 35 means drought. Here the monsoon average is 9.

INDICES

Combining greenness and heat: VHI



Simulated region. Here 10% of the area is in severe or extreme drought, 21% moderate or mild.

TCI: like VCI, but for temperature; hotter than usual scores lower.

VHI: the average of the two. Below 40 is drought; below 10 extreme.

INDICES

Which index for which drought

Index	Based on	Drought type
SPI, SPEI	Rainfall (and evaporation)	Meteorological; long windows for hydrological
NDVI anomaly, VCI	Greenness	Agricultural
TCI, Evaporative Stress Index	Temperature, water use	Agricultural, early warning
VHI	Greenness and temperature	Agricultural
Soil moisture anomaly	Top-soil water	Agricultural, early
Water storage anomaly	GRACE gravity	Groundwater, hydrological

No single index is enough: drought services combine several, then check them on the ground.

04

PART 4

Case studies and services

Marathwada, the Horn of Africa, California, Europe, and who monitors drought

CASE STUDY

Marathwada, Maharashtra, 2015–16

3 years
Of failed or weak monsoons in a row

April 2016
A water train starts carrying drinking water to Latur

5 lakh L
Drinking water per trip: 10 wagons of 50,000 litres

≈ 24 crore L
Delivered by the end of July 2016

- Meteorological drought became agricultural, then hydrological and groundwater drought.
- Reservoirs and most borewells in Latur ran dry.
- Rainfall and vegetation indices had shown the drought for many months before.

Sources: news reports (ABP Live, Al Jazeera, April 2016); Indian Railways.

CASE STUDY

Horn of Africa, 2020–2023

5
Failed rainy seasons in a row, late 2020 to early 2023

40 years
The worst drought in four decades

3
Countries worst hit: Ethiopia, Kenya and Somalia

Millions
People who needed food and water aid

Satellite rainfall and vegetation data tracked each failed season as it happened.

Each failed season added to the damage: livestock, crops and water sources were lost.

Attribution studies found climate change made the drought far more likely.

Sources: WMO; World Weather Attribution (2023), reported by Carbon Brief.

CASE STUDY

California’s Central Valley, 2012–2016

≈ 40 km³
Groundwater lost from the Central Valley, seen by GRACE

≈ 10 km³/yr
Rate of loss during the drought

33 cm
Ground sank near Corcoran in 8 months of 2015

2012–16
One of the most severe droughts in the state’s record

Farmers pumped groundwater to replace missing river water.

GRACE saw the lost mass from space; InSAR saw the ground sink.

The same drought links this lecture to land subsidence (Lecture 6).

Sources: Xiao et al. (2017), Geophysical Research Letters; NASA JPL (2015).

CASE STUDY

Europe, summer 2022

500 years
Possibly the worst drought in at least five centuries

47%
Of Europe under ‘warning’: clear soil-moisture deficit (August)

17%
Under ‘alert’: vegetation already affected

−16%
Grain maize yield against the five-year average

Rhine water levels fell so low that barges carried a fraction of their usual load.

Hydropower and power-station cooling were cut back.

The European Drought Observatory combined rainfall, soil moisture and vegetation data.

Sources: European Commission Joint Research Centre, European Drought Observatory (2022).

SERVICES

Who monitors drought

India

IMD rainfall and SPI; NADAMS agricultural drought maps from MNCFC every fortnight for 14 states; the national Drought Manual.

Copernicus drought observatories

European (EDO) and Global (GDO): combined indicators, free maps and data.

FEWS NET

Famine early warning for Africa, Central America and Asia, built on satellite rain and vegetation.

US Drought Monitor

Weekly map combining many indices with expert judgement.

05

PART 5

Engineering practice

Water decisions, a monitoring workflow, limits and exercises

ENGINEERING

Engineering decisions that need drought data

Reservoir operation

When to start rationing, and how much to release for irrigation and towns.

Water supply planning

Design storage and backup sources for the drought of record.

Irrigation

Schedule and allocate canal water where crops are most stressed.

Crop insurance and relief

Satellite indices support claims and drought declarations.

Energy and transport

Hydropower, power-plant cooling and river navigation.

Groundwater

Limit pumping before aquifers and the ground above them fail.

ENGINEERING

A drought monitoring workflow

01
Region and period (decades)

02
Rainfall: CHIRPS → SPI

03
NDVI: MODIS → VCI

04
Temperature → TCI → VHI

05
Soil moisture anomaly

06
Combine and classify

07
Check with crops, reservoirs, wells

08
Report by district

Tools: Google Earth Engine has CHIRPS, MODIS and SMAP ready to use.

Masks: crop masks, and remove irrigated land where greenness hides drought.

ENGINEERING

Limits and good practice

Limits

- Clouds hide crops in the monsoon (optical)
- Irrigation keeps fields green during drought
- Coarse soil moisture and GRACE data
- Short records for some sensors

Good practice

- Combine rainfall, vegetation, temperature and soil moisture
- Use crop and irrigation masks
- Use long baselines for ‘normal’
- Check against field reports, reservoirs and wells

EXERCISES

Try these

01 A crop reflects 45% in the near-infrared and 8% in red. What is its NDVI? And for 30% NIR and 12% red?

02 For this fortnight, NDVI over 20 years ranged from 0.30 to 0.70. Today it is 0.42. What is the VCI? Is it drought?

03 A district has VCI 30 and TCI 18. What is its VHI, and which drought class is that?

04 This season’s SPI-3 is −1.7. What class is that, and roughly how often is a season this dry or drier?

EXERCISES

Answers

01 (0.45 − 0.08) ÷ (0.45 + 0.08) ≈ **0.70**; (0.30 − 0.12) ÷ (0.30 + 0.12) ≈ **0.43**.

02 (0.42 − 0.30) ÷ (0.70 − 0.30) × 100 = **30**: below about 35, so drought.

03 (30 + 18) ÷ 2 = **24**: moderate drought (20–30).

04 **Severe drought** (−1.5 to −2). A season at −1.5 or drier happens about 7% of the time: roughly once in 15 years.

RECAP

Key points

01 Drought is a shortage of water relative to normal, and it spreads slowly.

02 It moves from rain to soil, rivers and groundwater, with growing delays.

03 Satellites measure rain, greenness, heat, soil moisture and water storage.

04 SPI, VCI, TCI and VHI compare each place with its own history.

05 Combine indices, check on the ground, and act before supplies run out.

FURTHER READING

References

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Thank you. Questions?

Figures and the animation were made for this lecture. Rainfall, NDVI, index maps and water-storage series are simulated, and labelled as such; the reflectance spectra are schematic. Index definitions after McKee et al. (1993) and Kogan (1995). Case-study facts: news reports and Indian Railways (Marathwada); WMO and World Weather Attribution (Horn of Africa); Xiao et al. (2017) and NASA JPL (California); European Commission JRC (Europe). Indian services: IMD, MNCFC (NADAMS), Government of India Drought Manual. Full references on the previous slide.